

PHY101 FINAL TERM PREPARATION

BY BRAINY SQUAD 100+ MCQ

1. How fast should you move away from a $6.0 \times 10^{14} \text{ Hz}$ light source to observe waves with a frequency of $4.0 \times 10^{14} \text{ Hz}$?

- A. $0.20c$
 - B. $0.38c$
 - C. $0.45c$
 - D. **$0.51c$**
 - E. $0.76c$
-

2. The units of the electric field are:

- A. $\text{J}/(\text{C} \cdot \text{m})$
 - B. **J/C**
 - C. $\text{J} \cdot \text{C}$
 - D. J/m
-

3. If two light waves are coherent:

- A. their amplitudes are the same
 - B. **their frequencies are the same**
 - C. their wavelengths are the same
 - D. their phase difference is constant
-

4. A 60-watt light bulb carries a current of 0.5 A. The total charge passing through it in one hour is:

- A. 120 C
 - B. 3600 C
 - C. **3000 C**
 - D. 2400 C
 - E. 1800 C
-

5. If the charge on a parallel-plate capacitor is doubled:

- A. the capacitance is halved
 - B. the capacitance is doubled
 - C. the electric field is halved
 - D. **the electric field is doubled**
-

Question No:6

When an object (iron/tungsten) is heated, it emits light. It emits infrared (IR) light when it is heated at a temperature _____.

- Below 800 degree C
 - **Between 800 degree C to 1600 degree C**
 - Between 1600 degree C to 2500 degree C
 - Above 2500 degree C
-

Question No:7

The reason there are two slits, rather than one, in a Young's experiment is:

- **To create a path length difference**
 - To increase the intensity
 - One slit is for frequency, the other for wavelength
 - One slit is for Electric fields, the other is for Magnetic fields
-

Question No:8

An electron is moving at $0.6c$. If we calculate its kinetic energy using $(1/2)mv^2$, we get a result that is:

- Just right
 - Just half enough
 - Twice the correct value
 - **About 28% too low**
-

Question No:9

During a slow adiabatic expansion of a gas:

- The pressure remains constant
 - Energy is added as heat
 - Work is done on the gas
 - **No energy enters or leaves as heat**
-

Question No:10

The theoretical upper limit for the frequency of electromagnetic waves is:

- Just slightly greater than that of red light
 - **There is no upper limit**
-

Question No:11

One revolution is the same as:

- 1 rad
 - 57 rad
 - $\pi/2$ rad
 - π rad
 - **2π rad**
-

Question No:12

For a body to be in equilibrium under the combined action of several forces:

- **All of the forces form pairs of equal and opposite forces**
 - All the forces must be applied at the same point
 - Any two of these forces must be balanced by a third force
-

Question No:13

Capacitors C_1 and C_2 are connected in series. The equivalent capacitance is given by:

- $\frac{C_1 C_2}{C_1 + C_2}$
 - $\frac{C_1 + C_2}{C_1 C_2}$
 - $\frac{1}{C_1} + \frac{1}{C_2}$
 - $\frac{C_1}{C_2}$
-

Question No:14

If the potential difference across a resistor is doubled:

- **Only the current is doubled**
 - Only the current is halved
 - Only the resistance is doubled
 - Only the resistance is halved
-

Question No:15

By using only two resistors, R_1 and R_2 , a student is able to obtain resistances of 3Ω , 4Ω , 12Ω , 3Ω , 4Ω , 12Ω , and 16Ω . The values of R_1 and R_2 (in ohms) are:

- 3, 43
 - 2, 122
 - **4, 124**
 - 3, 163
-

Question No:16

Faraday's law states that an induced emf is proportional to:

- The rate of change of the electric field
 - **The rate of change of the magnetic flux**
 - The rate of change of the electric flux
 - The rate of change of the magnetic field
-

17. Read the statement carefully and choose 'inference' from the given options.

If you go to a Mexican restaurant, you can infer that:

- ► They will not serve tea.
 - ► They will have hot dogs on the menu.
 - ► **They have specialized in Mexican food. (Correct Answer)**
 - ► They will have the best food in the town.
-

Question No:18

Which of the following electromagnetic radiations has photons with the greatest energy?

- Blue light
- Yellow light
- **X-rays**
- Radio waves

Question No:19

We desire to make an LC circuit that oscillates at 100 Hz using an inductance of 2.5H. We also need a capacitance of:

- 1 F
- 1 mF
- 1 μ F**
- 100 μ F

Question No:20

The wavelength of red light is 700 nm. Its frequency is:

- 4.30 * 10⁴ Hertz
- 4.30 * 10³ Hertz
- 4.30 * 10⁵ Hertz
- 4.30 * 10¹⁴ Hertz**

Question No:21

Which of the following statements is NOT TRUE about electromagnetic waves?

Electromagnetic waves satisfy Maxwell's Equation.

Electromagnetic waves cannot travel through space.

The reception of electromagnetic waves requires an antenna.

The electromagnetic radiation from a burning candle is unpolarized.

Question No:22

Radio waves and light waves are:

Longitudinal waves

Transverse waves

Electromagnetic and transverse both

Question No:23

An ideal gas expands into a vacuum in a rigid vessel. As a result, there is:

A change in entropy

A decrease of internal energy

An increase of pressure

A change in temperature

Question No:24

The Stern-Gerlach experiment makes use of:

A strong uniform magnetic field

A strong non-uniform magnetic field

A strong uniform electric field

A strong non-uniform electric field

Question No:25

A net torque applied to a rigid object always tends to produce:

Linear acceleration

Rotational equilibrium

Angular acceleration

Rotational inertia

Question No:26

As the wavelength of a wave in a uniform medium increases, its speed will:

Decrease

Increase

Remain the same

None of these

27. The electric field between two oppositely charged plates is:

- ► **Uniform**
- ► Non-uniform
- ► Zero
- ► Varying

28. The total charge on a capacitor depends on:

- ► Its size
- ► The voltage across it
- ► **Both its size and the voltage across it**
- ► The type of material used for the dielectric

29. In a capacitor, when the distance between the plates is doubled while keeping the charge constant, the potential difference between the plates:

- ► Remains the same
- ► **Doubles**
- ► Halves
- ► Becomes zero

30. A capacitor is connected to a battery and fully charged. If the battery is removed and the capacitor is disconnected from the circuit, the charge on the capacitor:

- ► Decreases
- ► Increases
- ► **Remains the same**
- ► Becomes zero

31. The capacitance of a parallel plate capacitor is given by the formula:

- ► **$C = \epsilon_0 A / d$**
- ► $C = \epsilon_0 A \times d$
- ► $C = \epsilon_0 \times d / A$
- ► $C = A / (\epsilon_0 \times d)$

32. Which of the following is true for a dielectric material inserted into a capacitor?

- ► **It increases the capacitance**
- ► It decreases the capacitance
- ► It has no effect on the capacitance
- ► It changes the charge on the capacitor

33. A spherical conductor has a net charge of +Q. The electric field inside the conductor is:

- ► Non-zero
- ► **Zero**
- ► Positive
- ► Negative

34. If a dielectric material is inserted into a parallel-plate capacitor, the electric field between the plates:

- ► **Decreases**
- ► Increases
- ► Remains the same
- ► Becomes zero

35. The unit of electric flux is:

- ► **$\text{N}\cdot\text{m}^2/\text{C}$**
- ► $\text{C}\cdot\text{m}^2/\text{N}$
- ► N/C
- ► $\text{C}\cdot\text{N}/\text{m}$

36. The electric flux through a Gaussian surface depends on:

- ► **The charge enclosed within the surface**
- ► The size of the surface
- ► The material of the surface
- ► The potential difference across the surface

37. The electric potential at a point due to a point charge is given by the formula:

- ► **$V = kQ / r$**
- ► $V = kQ r$
- ► $V = kQ / r^2$
- ► $V = kQ r^2$

38. The potential difference across a conductor is proportional to the:

- ► **Current passing through it**
- ► Resistance of the conductor
- ► Power dissipated by the conductor
- ► Charge on the conductor

39. If the resistance of a conductor is doubled, the power dissipated in it is:

- ► **Halved**
- ► Doubled
- ► Remains the same
- ► Quadrupled

40. The resistance of a conductor is directly proportional to:

- ► Its length
- ► **Its length and inversely proportional to its cross-sectional area**
- ► Its cross-sectional area
- ► Its temperature only

41. The SI unit of electric current is:

- ► Coulomb
- ► Ampere
- ► **Ampere**
- ► Volt

42. In Ohm's law, the relationship between voltage, current, and resistance is given by:

- ► **$V = I \times R$**
- ► $V = I / R$
- ► $V = R / I$
- ► $V = I + R$

43. The total resistance of resistors connected in series is:

- ► **The sum of the individual resistances**
- ► The reciprocal of the sum of the individual resistances
- ► The product of the individual resistances
- ► None of the above

44. The total resistance of resistors connected in parallel is:

- ► The sum of the individual resistances
- ► **The reciprocal of the sum of the reciprocals of the individual resistances**
- ► The product of the individual resistances
- ► None of the above

45. The equivalent resistance of two resistors connected in series is:

- ► **$R_1 + R_2$**
- ► $R_1 \times R_2$
- ► $(R_1 + R_2)/2$
- ► None of the above

46. The equivalent resistance of two resistors connected in parallel is:

- ► **$1 / (1/R_1 + 1/R_2)$**
- ► $R_1 + R_2$
- ► $(R_1 \times R_2) / (R_1 + R_2)$

- ► None of the above

47. The power dissipated in a resistor is given by:

- ► $P = I^2 R$
- ► $P = I / R$
- ► $P = I \times R$
- ► $P = R^2 I$

48. The rate of change of electric flux is equal to the:

- ► **Induced emf**
- ► Electric field
- ► Current
- ► Voltage

49. The magnetic field produced by a current-carrying conductor is:

- ► **Circular and concentric around the conductor**
- ► Linear and parallel to the conductor
- ► Radial and perpendicular to the conductor
- ► Uniform in all directions

50. The force on a charged particle moving in a magnetic field depends on:

- ► The velocity of the particle
- ► The magnetic field strength
- ► The angle between the velocity and magnetic field
- ► **All of the above**

51. The magnetic force on a current-carrying wire is given by:

- ► $F = BIL \sin\theta$
- ► $F = BIL \cos\theta$
- ► $F = B / IL$
- ► $F = BIL$

52. Faraday's law of induction states that:

- ► **The induced emf is proportional to the rate of change of magnetic flux**
- ► The induced emf is proportional to the magnetic field strength
- ► The induced emf is independent of the rate of change of magnetic flux
- ► The induced emf is directly proportional to the electric field

53. The unit of magnetic flux is:

- ► **Weber (Wb)**
- ► Tesla (T)
- ► Gauss (G)
- ► Newton (N)

54. A changing magnetic field induces an electric field according to:

- ► **Faraday's law of induction**
- ► Gauss's law
- ► Ampère's law
- ► Biot-Savart law

55. A solenoid produces a magnetic field that is:

- ► **Uniform inside and weaker outside**
- ► Non-uniform inside and stronger outside
- ► Uniform both inside and outside
- ► Weak inside and stronger outside

56. The number of significant figures in 0.00150 is:

- ► **3**
- ► 4
- ► 5
- ► 2

57. For a body to be in equilibrium under the combined action of several forces:

- ► All the forces must be applied at the same point
- ► All of the forces form pairs of equal and opposite forces
- ► Any two of these forces must be balanced by a third force
- ► **The sum of the torques about any point must equal zero**

58. A bucket of water is pushed from left to right with increasing speed across a horizontal surface. Consider the pressure at two points at the same level in the water:

- ► It is the same
- ► It is higher at the point on the left
- ► **It is higher at the point on the right**
- ► At first, it is higher at the point on the left but as the bucket speeds up it is lower there

59. An organ pipe with both ends open is 0.85m long. Assuming that the speed of sound is 340m/s, the frequency of the third harmonic of this pipe is:

- ► 400 Hz
- ► **200 Hz**
- ► 300 Hz
- ► 600 Hz

60. Capacitors C_1 and C_2 are connected in series. The equivalent capacitance is given by:

- ► **$C_1 C_2 / (C_1 + C_2)$**
- ► $(C_1 + C_2) / C_1 C_2$
- ► $1 / (C_1 + C_2)$
- ► C_1 / C_2

61. If the potential difference across a resistor is doubled:

- ► Only the current is doubled
- ► Only the current is halved
- ► Only the resistance is doubled
- ► **Only the resistance is halved**

62. By using only two resistors, R_1 and R_2 , a student is able to obtain resistances of $3\ \Omega$, $4\ \Omega$, $12\ \Omega$, and $16\ \Omega$. The values of R_1 and R_2 (in ohms) are:

- ► **3, 4**
- ► 2, 12
- ► 3, 16
- ► 4, 12

63. Faraday's law states that an induced emf is proportional to:

- ► The rate of change of the electric field
- ► **The rate of change of the magnetic flux**
- ► The rate of change of the electric flux
- ► The rate of change of the magnetic field

64. A generator supplies 100V to the primary coil of a transformer. The primary has 50 turns and the secondary has 500 turns. The secondary voltage is:

- ► **1000V**
- ► 500V
- ► 250V
- ► 100V

65. Which of the following electromagnetic radiations has photons with the greatest energy?

- ► **X-rays**
- ► Visible light
- ► Radio waves
- ► Microwaves

66. In a magnetic field, a moving charged particle experiences a force. The direction of this force depends on:

- ► The velocity of the particle
- ► The charge of the particle
- ► **Both the velocity of the particle and the direction of the magnetic field**
- ► Only the charge of the particle

67. "The sum of the currents into a junction equals the sum of the currents out of the junction" is a consequence of:

- ► Newton's third law
- ► Ohm's law
- ► Newton's second law
- ► Conservation of energy
- ► **Conservation of charge**

68. "The sum of the emf's and potential differences around a closed loop equals zero" is a consequence of:

- ► Newton's third law
- ► Ohm's law
- ► Newton's second law
- ► Conservation of energy
- ► **Conservation of charge**

69. Evidence for the wave nature of matter is:

- ► **Electron diffraction experiments of Davisson and Germer**
- ► Thompson's measurement of e/m
- ► Young's double slit experiment
- ► The Compton effect

70. A conducting sphere with radius R is charged until the magnitude of the electric field just outside its surface is E . The electric potential of the sphere, relative to the potential far away, is:

- ► Zero
- ► E/R
- ► E/R^2
- ► **ER**

71. Radio waves and light waves are:

- ► **Electromagnetic and transverse**

72. An ideal gas expands into a vacuum in a rigid vessel. As a result there is:

- ► **A change in entropy**
- ► An increase of pressure
- ► A change in temperature
- ► A decrease of internal energy

73. Capacitors A and B are identical. Capacitor A is charged so it stores 4 J of energy and capacitor B is uncharged. The capacitors are then connected in parallel. The total stored energy in the capacitors is now:

- ► 16 J
- ► 8 J
- ► 4 J
- ► **2 J**

74. Capacitors C_1 and C_2 are connected in parallel. The equivalent capacitance is given by:

- ► $C_1 C_2 / (C_1 + C_2)$
- ► $(C_1 + C_2) / C_1 C_2$
- ► $1 / (C_1 + C_2)$
- ► C_1 / C_2
- ► **$C_1 + C_2$**

75. Capacitors C_1 and C_2 are connected in series. The equivalent capacitance is given by:

- ► $C_1 C_2 / (C_1 + C_2)$
- ► $(C_1 + C_2) / C_1 C_2$
- ► $1 / (C_1 + C_2)$
- ► C_1 / C_2
- ► **$C_1 + C_2$**

76. During an adiabatic process, an object does 100 J of work and its temperature decreases by 5 K. During another process, it does 25 J of work and its temperature decreases by 5 K. Its heat capacity for the second process is:

- ► **20 J/K**
- ► 24 J/K
- ► 5 J/K
- ► 15 J/K

77. A virtual image is one:

- ► **Toward which light rays converge but do not pass through**
- ► From which light rays diverge but do not pass through
- ► From which light rays diverge as they pass through
- ► Toward which light rays converge and pass through

78. A generator supplies 100 V to the primary coil of a transformer. The primary has 50 turns and the secondary has 500 turns. The secondary voltage is:

- ► **1000 V**
- ► 500 V
- ► 250 V
- ► 100 V

79. The rate at which electrical energy is used may be measured in:

- ► Watt/second
- ► Watt·second
- ► **Watt**
- ► Joule·second

80. The units of the Planck constant h are those of:

- A. energy
- B. power
- C. momentum
- **D. angular momentum**
- E. frequency

81. Which of the following is true of all virtual images?

- A. They can be seen but not photographed
- B. They are ephemeral
- C. They are smaller than the objects
- D. They are larger than the objects
- E. **None of the above**

82. In constructing a thermometer it is NECESSARY to use a substance that:

- A. **expands with rising temperature**
- B. expands linearly with rising temperature
- C. will not freeze
- D. will not boil
- E. undergoes some change when heated or cooled

83. A 60-watt light bulb carries a current of 0.5 A. The total charge passing through it in one hour is:

- A. 120 C
- B. **3600 C**
- C. 3000 C
- D. 2400 C
- E. 1800 C

84. If the charge on a parallel-plate capacitor is doubled:

- A. the capacitance is halved
- B. the capacitance is doubled
- C. the electric field is halved
- D. **the electric field is doubled**

85. The wavelength of red light is 700 nm. Its frequency is:

- A. 4.30×10^4 Hertz
- B. **4.30×10^5 Hertz**
- C. 4.30×10^3 Hertz
- D. 4.30×10^2 Hertz

86. A capacitor C “has a charge Q”. The actual charges on its plates are:

- A. **Q, Q**
- B. $Q/2$, $Q/2$
- C. **Q, -Q**
- D. $Q/2$, $-Q/2$

87. How fast should you move away from a 6.0×10^{14} Hz light source to observe waves with a frequency of 4.0×10^{14} Hz?

- A. 0.20c
- B. 0.38c
- C. 0.45c
- D. 0.51c
- E. **0.76c**

88. The units of the electric field are:

- A. $\text{J}/(\text{C}\cdot\text{m})$
- B. J/C
- C. $\text{J}\cdot\text{C}$
- D. J/m

89. If two light waves are coherent:

- A. their amplitudes are the same
- B. their frequencies are the same
- C. their wavelengths are the same
- D. **their phase difference is constant**

90. In a photoelectric effect experiment at a frequency above cut off, the stopping potential is proportional to:

- A. the energy of the least energetic electron before it is ejected
- B. the energy of the least energetic electron after it is ejected
- C. the energy of the most energetic electron before it is ejected
- D. **the energy of the most energetic electron after it is ejected**

91. The total energy in an LC circuit is $5.0 \times 10^{-6} \text{ J}$. If $L = 25 \text{ mH}$ the maximum current is:

- A. 10 mA
- B. 14 mA
- C. **20 mA**
- D. 28 mA

92. The quantum number m_s is most closely associated with what property of the electron in an atom?

- A. Magnitude of the orbital angular momentum
- B. Energy
- C. **z component of the spin angular momentum**
- D. z component of the orbital angular momentum

93. If 500 J of work are required to carry a charged particle between two points with a potential difference of 20 V, the magnitude of the charge on the particle is:

- A. **0.040 C**
- B. 12.5 C
- C. 20 C
- D. cannot be computed unless the path is given

94. The smallest particle of any chemical element that can exist by itself and yet retain the qualities that distinguish it as that element is:

- A. an electron
- B. a proton
- C. a neutron
- D. **an atom**
- E. a molecule

95. A particle with a charge of $5.5 \times 10^{-8} \text{ C}$ is 3.5 cm from a particle with a charge of $-2.3 \times 10^{-8} \text{ C}$. The potential energy of this two-particle system, relative to the potential energy at infinite separation, is:

- A. $3.2 \times 10^{-4} \text{ J}$
- B. $-3.2 \times 10^{-4} \text{ J}$
- C. $9.3 \times 10^{-3} \text{ J}$
- D. $-9.3 \times 10^{-3} \text{ J}$
- E. zero

96. A farad is the same as a:

- A. J/V
- B. V/J
- C. C/V
- D. V/C
- E. N/C

97.1 weber is the same as:

- A. 1 V/s
- B. 1 T/s
- C. 1 T/m
- D. $1 \text{ T} \cdot \text{m}^2$

98. Coherent sources of light mean:

- A. They must have same speeds
- B. They must have same frequency
- C. They must have same amplitude
- D. They must have fixed phase relative to each other

99. Room temperature is about 20 degrees on the:

- A. Kelvin scale
- B. Celsius scale
- C. Fahrenheit scale
- D. absolute scale
- E. C major scale

100. The appearance of color in the thin film is due to:

- Interference

101. When you stand in front of a plane mirror, your image is:

- A. real, erect, and smaller than you
- B. real, erect, and the same size as you
- C. virtual, erect, and smaller than you
- D. virtual, erect, and the same size

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